

Chapter 1

Conclusion and Recommendations

In conclusion, a LoRa-based transmission and receiver system was built to emulate a real-world radiosonde weather balloon module. The overall system was split up into a transmitter module and a receiver module. The transmitter module was equipped with a battery module that allowed for wireless usage of the device, a sensor suite that could take atmospheric and GPS measurements, a LoRa module, and an OLED display and control module to navigate through various modes that aided in the testing process. The receiver module featured a LoRa module and a display and control module, also to help with navigating through the various LoRa modes.

Before testing, all ethical considerations were taken into account, as referenced by the ethics clearance as seen in figure ?? in appendix section ?. Once ethics clearance was obtained, the testing procedure could continue.

Once the overall system was fully designed and implemented with hardware, a full suite of testing procedures was developed that would aid in the determination of the best LoRa parameters for a radiosonde module being used in real life. Testing was performed on the rugby fields at UCT and then between Lions Head Mountain and the Tygerburg Nature Reserve, which ended up being a distance of roughly 20km. Once testing was completed, the data was analysed and it was determined that the best LoRa parameters were as described in table ?, since they give an overall well-rounded performance when compared to the other Lora parameters.

ATPs 1 and 3 have been met, with the transmission rates being defined in table ? in symbols/s, and with all of the components meeting the standard design minimum temperature of -40°C. ATP 2 has not been met, since the system was not tested to a distance of 100km due to physical restrictions in actually performing a line-of-sight test over that sort of distance.

It is recommended in order to gain a full understanding of the true effect of the LoRa parameters on the LoRa transmission performance, that a suitable long-range test is performed over a distance greater than 20km. Once a test greater than 20km has been performed, a better understanding will be had of the effect that the parameters have on the system for extremely far distances. Doing a test of this nature may require a trip out into the Karoo where line-of-sight tests can be performed, without any interference from any Fresnel Zones. Another recommendation is to use a combination of the GPS module and the pressure sensor to determine the altitude of the module for more accurate results than just the pressure sensor. It is also recommended that the LoRa modules used in further research have temperature-compensated oscillators, which would drastically help system performance especially at lower bandwidths, due to the temperature-compensated oscillators helping in mitigating the frequency drift that was seen in some of the testing as seen in figure ?. Another aspect which I would recommend for future work is that the same microcontroller units are implemented for the transmitter and receiver modules, as this would ease the development process since only one software toolchain would be required as opposed to two. The last recommendation is to stay away from serial-based sensors, and to

favour the I2C and SPI based sensors, as from my experience, reading and writing data from these sensors tends to be a smoother process.

Bibliography

Glossary